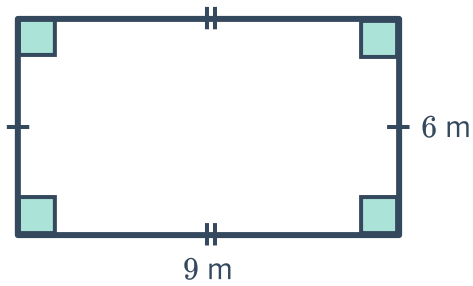


Areas of rectangles, squares and triangles

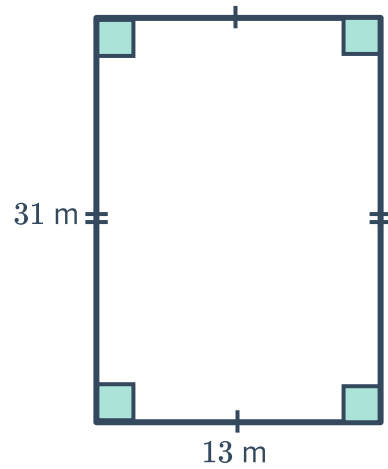


1 Find the area of the following rectangles:

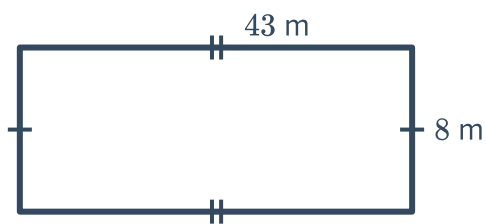
a



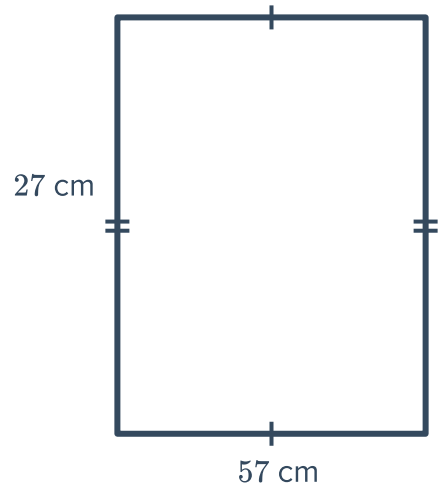
b



c



d

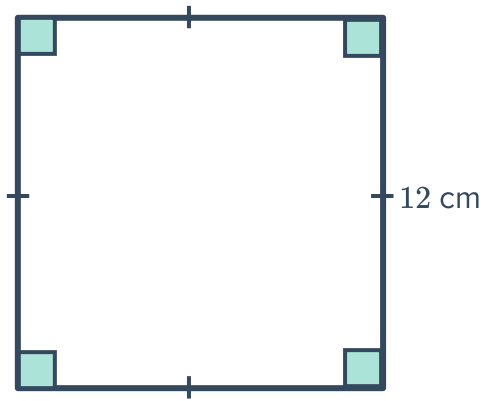


- 2 Find the area of a rectangle whose length is 12 cm and width is 5 cm.
- 3 Find the area of a rectangle whose length is 11 m and width is 8 m.
- 4 If the area of a rectangle is 99 cm^2 and its width is 9 cm, find its length.
- 5 If the area of a rectangle is 160 cm^2 and its length is 20 cm, find its width.

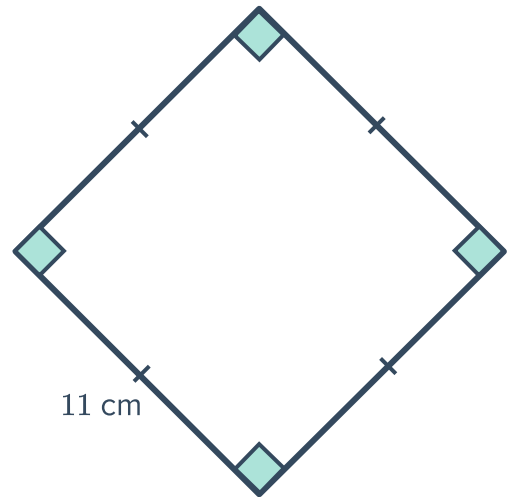
 **6** Find the area of the following squares:



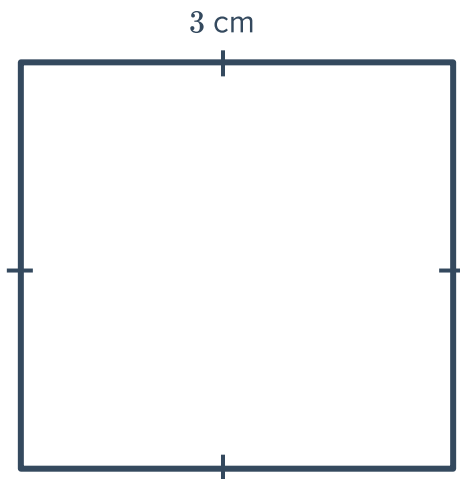
a



b



c



 **7** Find the length of each side of a square whose area is:

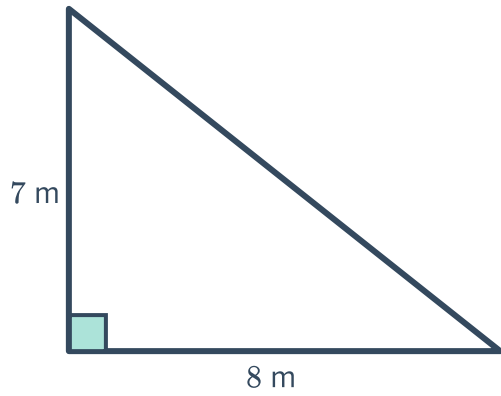
a 64 cm^2

b 36 m^2

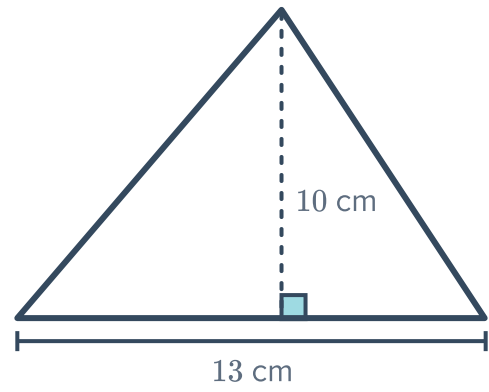
8 Find the area of the following triangles:



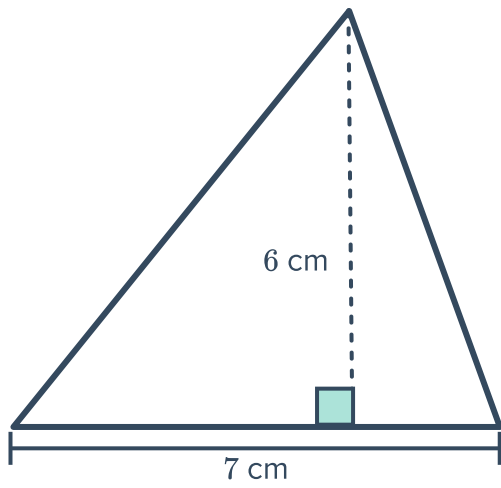
a



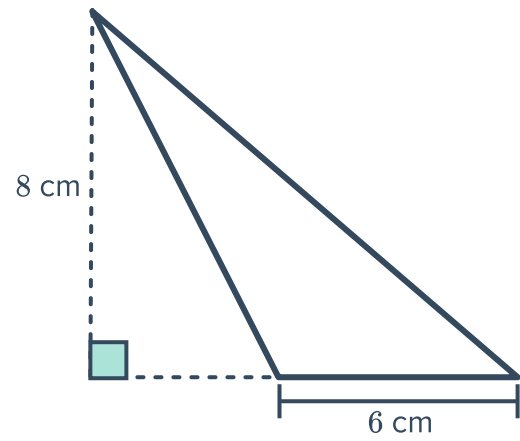
b



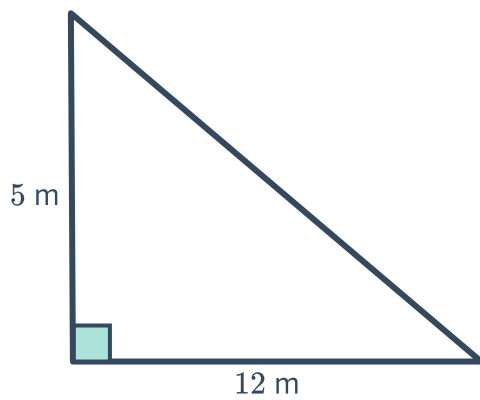
c



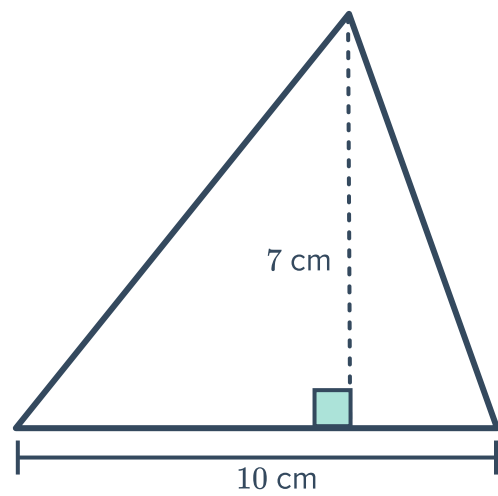
d



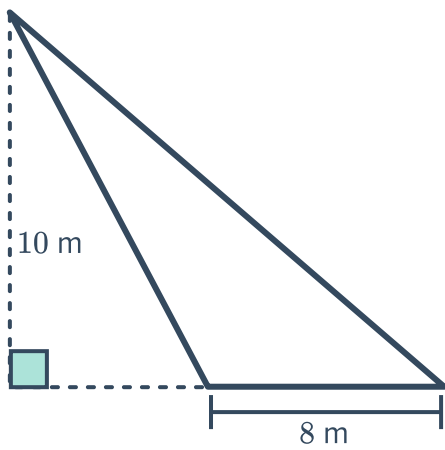
e



f



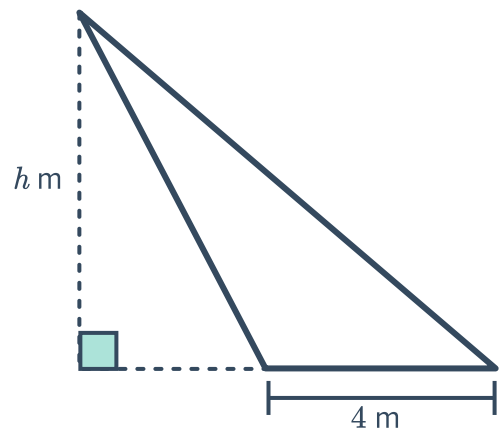
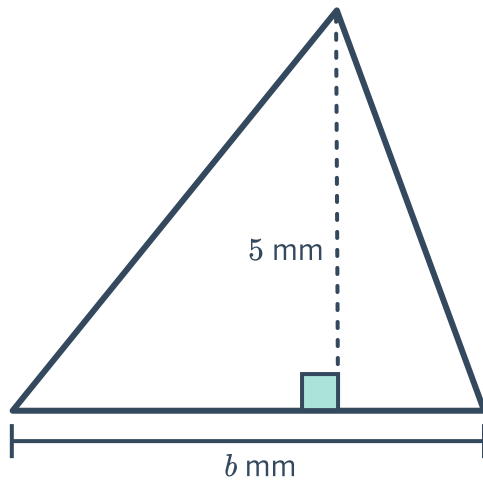
g



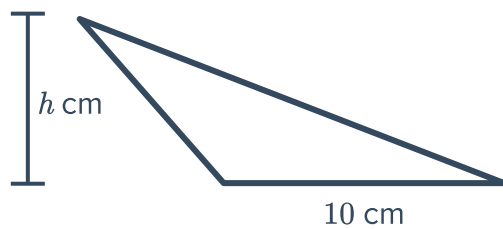
9 Find the value of the pronumeral for each of the following triangles, given the area:

a The area of this triangle is 20 mm^2

b The area of this triangle is 48 m^2 .



c The area of this triangle is 130 cm^2 .





Areas of parallelograms

11 Find the area of the following parallelograms:

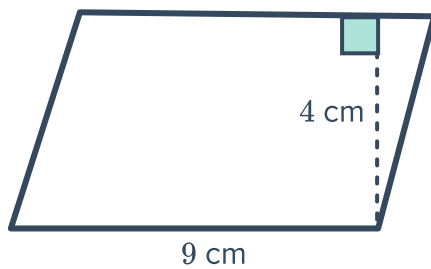
a



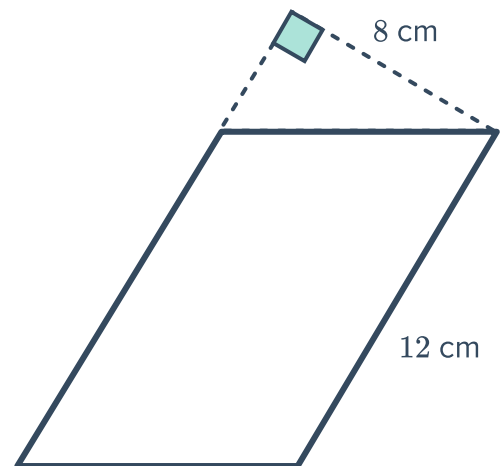
b



c



d



12 Find the area of a parallelogram with:

- a A base of 5 mm and perpendicular height of 2 mm.
- b A base of 15 cm and a perpendicular height of 7 cm.
- c A base of 10 cm and a perpendicular height 5 cm.

13 Find the base length of a parallelogram with area of 96 cm^2 and perpendicular height of 8 cm.

14 Find the height of a parallelogram with area of 84 cm^2 and perpendicular base length of 6 cm.

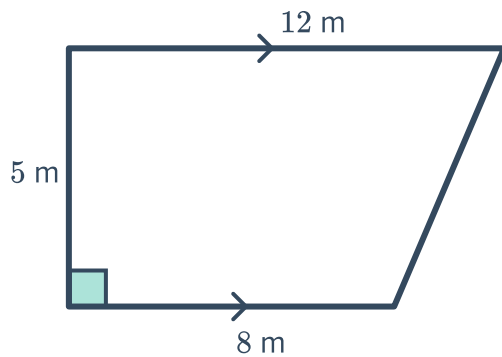


- 15** Find the perpendicular height, h , of a parallelogram that has an area of 45 cm^2 and a base length of 5 cm .
- 16** Find the base length, b , of a parallelogram that has an area of 216 mm^2 and a perpendicular height of 12 mm .

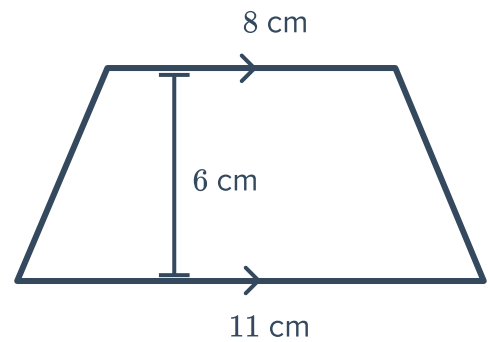
Areas of trapeziums

- 17** Find the area of the following trapeziums:

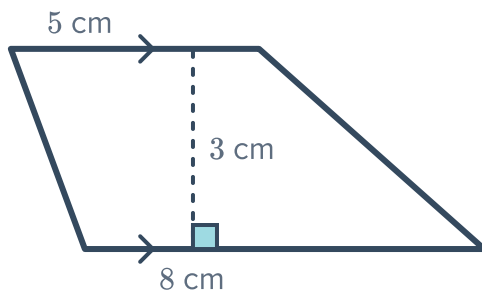
a



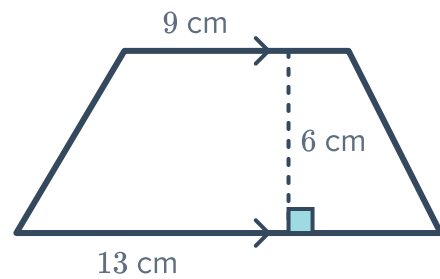
b



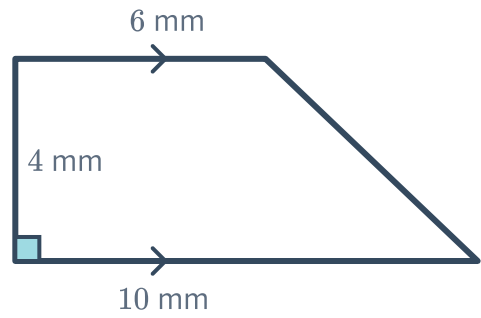
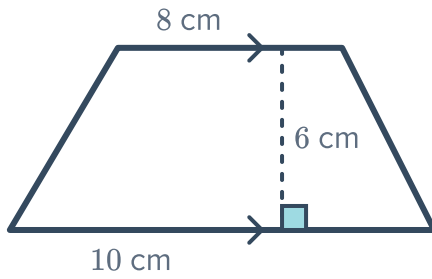
c



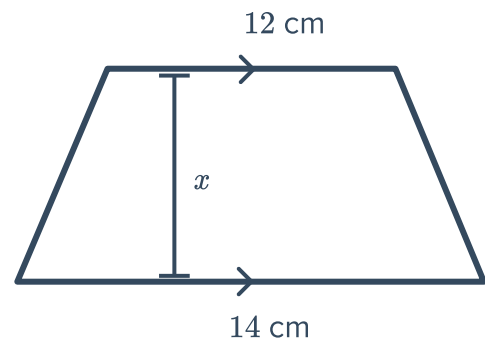
d



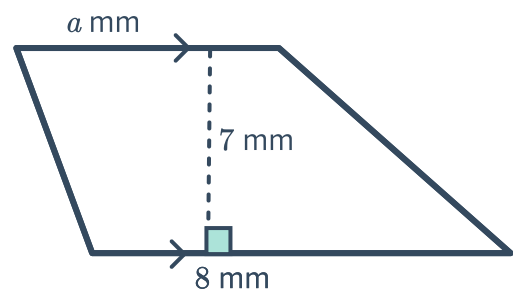
e



- 18 Find the value of x if the area of the trapezium shown is 65 cm^2 .



- 19 Find the value of a if the area of the trapezium is 42 mm^2 .

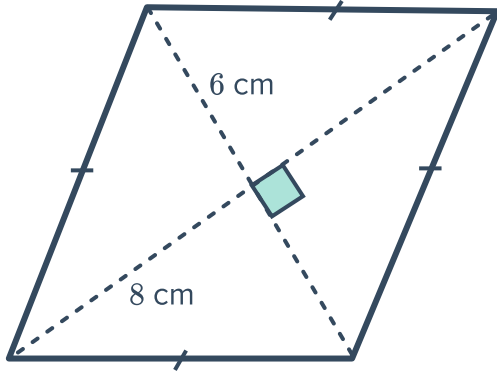




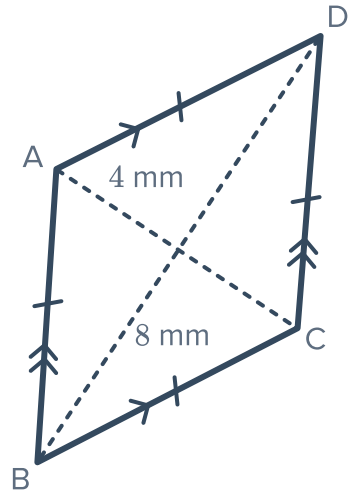
Areas of Kites and Rhombuses

 **20** Find the area of each rhombus below:

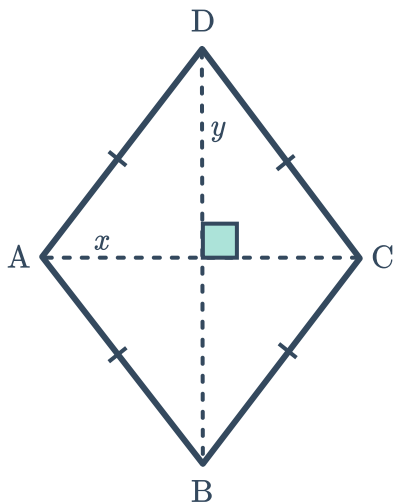
a



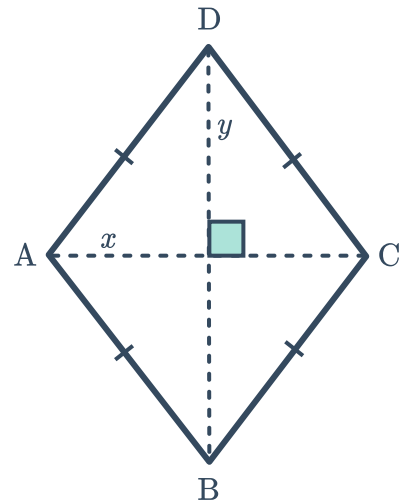
b



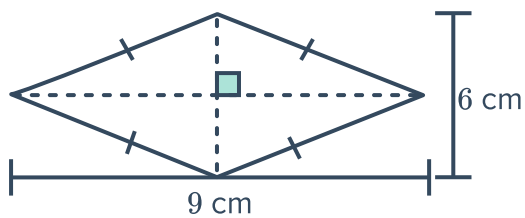
c $x = 10$ and $y = 12$



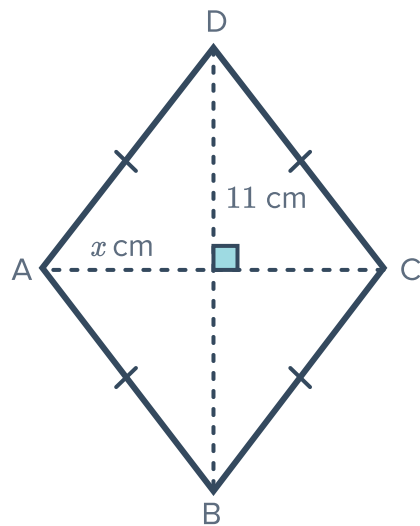
d $x = 4$ cm and $y = 8$ cm



e

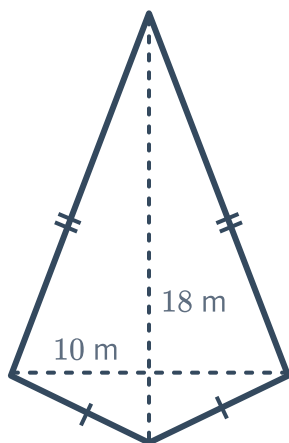


- 21** Rhombus $ABCD$ has an area of 55 cm^2 . If diagonal $BD = 11 \text{ cm}$, and $AC = x \text{ cm}$, find x .

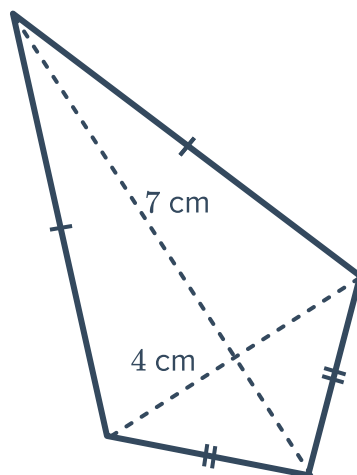


- 22** Find the area of the following kites:

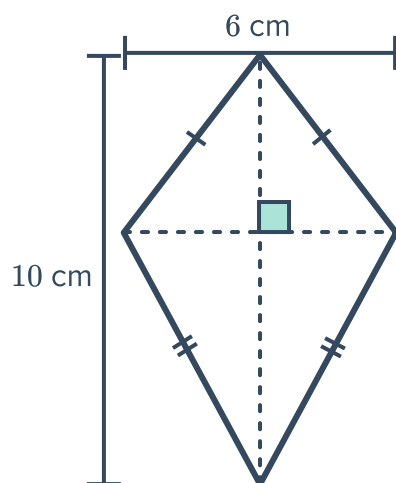
a



b

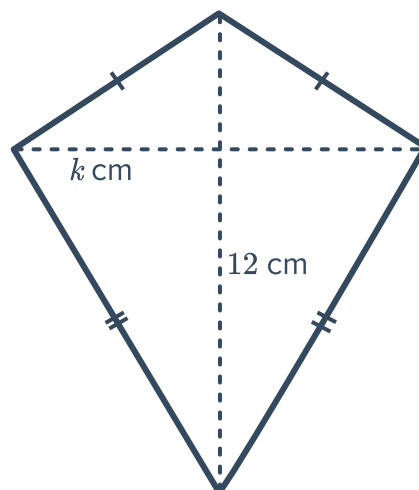


c



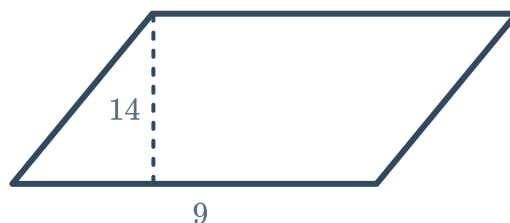
23 The area of a kite is 308 cm^2 and one of the diagonals is 47 cm . If the length of the other diagonal is $y \text{ cm}$, what is the exact value of y ?

24 The following kite has an area of 48 cm^2 .
The length of one of its diagonals is 12 cm .
Find the length of the other diagonal, k .

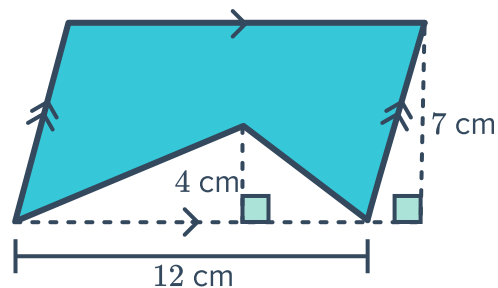


Applications

25 A rectangular sign casts a shadow on the ground as shown. What is the area of the shadow?

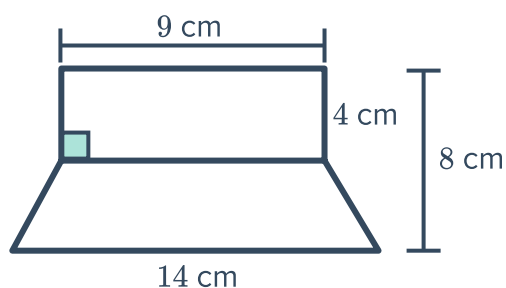


26 Find the shaded area shown in the figure:

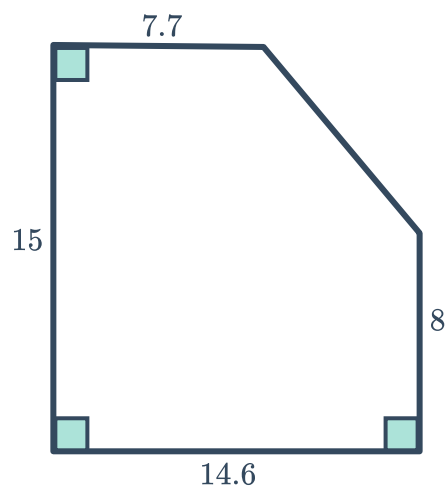


27 Sharon has purchased a rectangular piece of fabric measuring 12 m in length and 7 m in width. What is the area of the largest triangular piece she can cut out from it?

28 A rectangle is placed on top of a trapezium as shown. Find the area of the composite shape.



29 Find the area of the figure shown. All dimensions are in millimetres.



dimensions are in metres.

